

ROGER CREUS CASTANYER

PHD STUDENT



INFORMATION

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- 🌐 roger-creus.github.io
- 📍 Montreal, Quebec, Canada
- 📅 Expected PhD Graduation: Aug. 2028
- 🔗 Python / PyTorch / JAX / C++ / C#

EDUCATION

PROFILE

I am from Barcelona (Spain) and live in Montreal (Canada) as I pursue my PhD at Mila Quebec AI Institute and the University of Montreal, supervised by Professors *Pablo Samuel Castro* and *Glen Berseth*.

My research focuses on developing **intelligent agentic systems** that continually learn and develop skills to tackle increasingly complex problems.

Interests AI Agents, Reinforcement Learning, Foundation Models, LLMs / VLMs.

WORK EXPERIENCE



Research MSc in Computer Science & AI

Mila Quebec AI Institute and University of Montreal

Sep 2022 - Sep 2024

Member of the Robotics and Embodied AI lab (REAL), supervised by Professor Glen Berseth.

GPA: 4.225/4.3

PUBLICATIONS

2026

ARM-FM: Automated Reward Machines via Foundation Models for Compositional Reinforcement Learning
Roger Creus Castanyer, Faisal Mohamed, Pablo Samuel Castro, Cyrus Neary, Glen Berseth. [ICLR 2026](#)

PopuLoRA: Co-Evolving LLM Populations for Reasoning Self-Play

Roger Creus Castanyer, Geoffrey Bradway, Lorenz Wolf, Maxwill Lin, Augustine N. Mavor-Parker, Matthew James Sargent. *Under review*

Agentick: A Unified Benchmark for General Sequential Decision-Making Agents

Roger Creus Castanyer, Pablo Samuel Castro, Glen Berseth. *Under review*

unix-ctf: Procedural Environments for Unix-Competence Reinforcement Learning

Geoffrey Bradway, Roger Creus Castanyer, Lorenz Wolf, Maxwill Lin, Matthew James Sargent, Augustine N. Mavor-Parker. *Under review*

Breaking the Solver Bottleneck: Training Task Generators at the Learnable Frontier

Lorenz Wolf, Connor Watts, Roger Creus Castanyer, Geoffrey Bradway, Maxwill Lin, Augustine N. Mavor-Parker, Matthew Daborn-Sargent. *Under review*

2025

Stable Gradients for Stable Learning at Scale in Deep Reinforcement Learning

Roger Creus Castanyer, Obando, J., Lu Li, Pierre-Luc Bacon, Glen Berseth, Aaron Courville, Pablo S. Castro. [NeurIPS 2025](#) [Spotlight](#)

What Matters for Maximizing Data Reuse In Value-based Deep Reinforcement Learning

Roger Creus Castanyer, Glen Berseth, Pablo Samuel Castro. [Finding the Frame Workshop, RLC 2025](#)

2024

Improving Intrinsic Exploration by Creating Stationary Objectives

Roger Creus Castanyer, Joshua Romoff, Glen Berseth. [ICLR 2024](#)

Surprise-Adaptive Intrinsic Motivation for Unsupervised Reinforcement Learning

Adriana Hugessen (*), Roger Creus Castanyer (*), Faisal Mohamed (*), Glen Berseth. [RLC 2024 · IMOL Workshop, NeurIPS 2023](#) [Oral](#)

RLXplore: Accelerating Research in Intrinsically-Motivated Reinforcement Learning

Mingqi Yuan (*), Roger Creus Castanyer (*), Bo Li, X. Jin, Glen Berseth, W. Zeng. [TMLR 2024](#)

2021

Unsupervised Skill-Discovery and Skill-Learning in Minecraft

Juanjo Nieto, Roger Creus Castanyer, Xavier Giro-i-Nieto. [URL Workshop, ICML 2021](#)

Integration of Convolutional Neural Networks in Mobile Applications

Roger Creus Castanyer, Xavier Franch, Silverio Martinez. [AI Engineering Workshop, ICSE 2021](#)

Which Design Decisions in AI-enabled Mobile Applications Contribute to Greener AI?

Roger Creus Castanyer, Xavier Franch, Silverio Martinez. [Empirical Software Engineering Journal \(EMSE 2021\)](#)

SCHOLARSHIPS

- 2026** **Doctoral Research Scholarship (B2)** — Fonds de Recherche du Québec (FRQNT)
- 2025** **AI Research Scholarship** — University of Montreal
- 2024** **Academic Excellence Scholarship** — University of Montreal
- 2023** **End-of-studies Scholarship** — University of Montreal
- 2021** **Introduction to Research Scholarship (INIREC)** — Banco Santander, Spain

OTHER ACHIEVEMENTS

Contributor to Open-Source AI Software

Contributor to widely used AI repositories — Agentick, GlyphBench, CleanRL, RLLTE, RLeXplore, Wave-Defense RL, Blokus RL, Lux AI, and xgenius — all available on GitHub.

Content Creator in Artificial Intelligence

I run the YouTube channel [@whoami_AI](#) with educational AI content; some videos have reached 30k+ views.

Basketball Player — Fundació Privada Sagrada Família Claror | Sep. 2008 - Jun. 2020

Highest level played: the First Catalan League in the 2017–2018 and 2018–2019 seasons.

Basketball Coach — Fundació Privada Sagrada Família Claror | Sep. 2014 - Jun. 2022

Coached and mentored youth teams (ages 8–16) competing at a high level in multiple tournaments.
